ECONOMICS
Week 08
Breakeven Analysis

Sardar Shahbaz A. Naqvi

Break-Even Point Analysis

A decision-making aid that enables a manager to determine whether a particular volume of sales will result in **losses** or **profits**

Basic Concepts

Variable costs are costs that change with changes in production levels or sales.
Examples include: *Costs of materials used in the production of the goods*

Fixed costs remain roughly the same regardless of sales/output levels.
Examples include: *Rent, Insurance and Wages*

Revenue is the total income received.

Profit is the money you have after subtracting fixed and variable cost from revenue.

What it can be used for?

- Monthly expenses- use it to see if your income is more then your expenses
- Determine minimum price product can be sold for
- Determine optimum price product can be sold for
- Calculate effects of marketing programs on price

The Break-Even Point (BEP)

$$\underbrace{P(X)}_{\textit{Revenue}} = \underbrace{F + V(X)}_{\textit{Total Costs}}$$

Where:

- F = fixed costs
- V = variable costs per unit
- X = volume of output (in units)
- P = price per unit

The Break-Even Point (BEP)

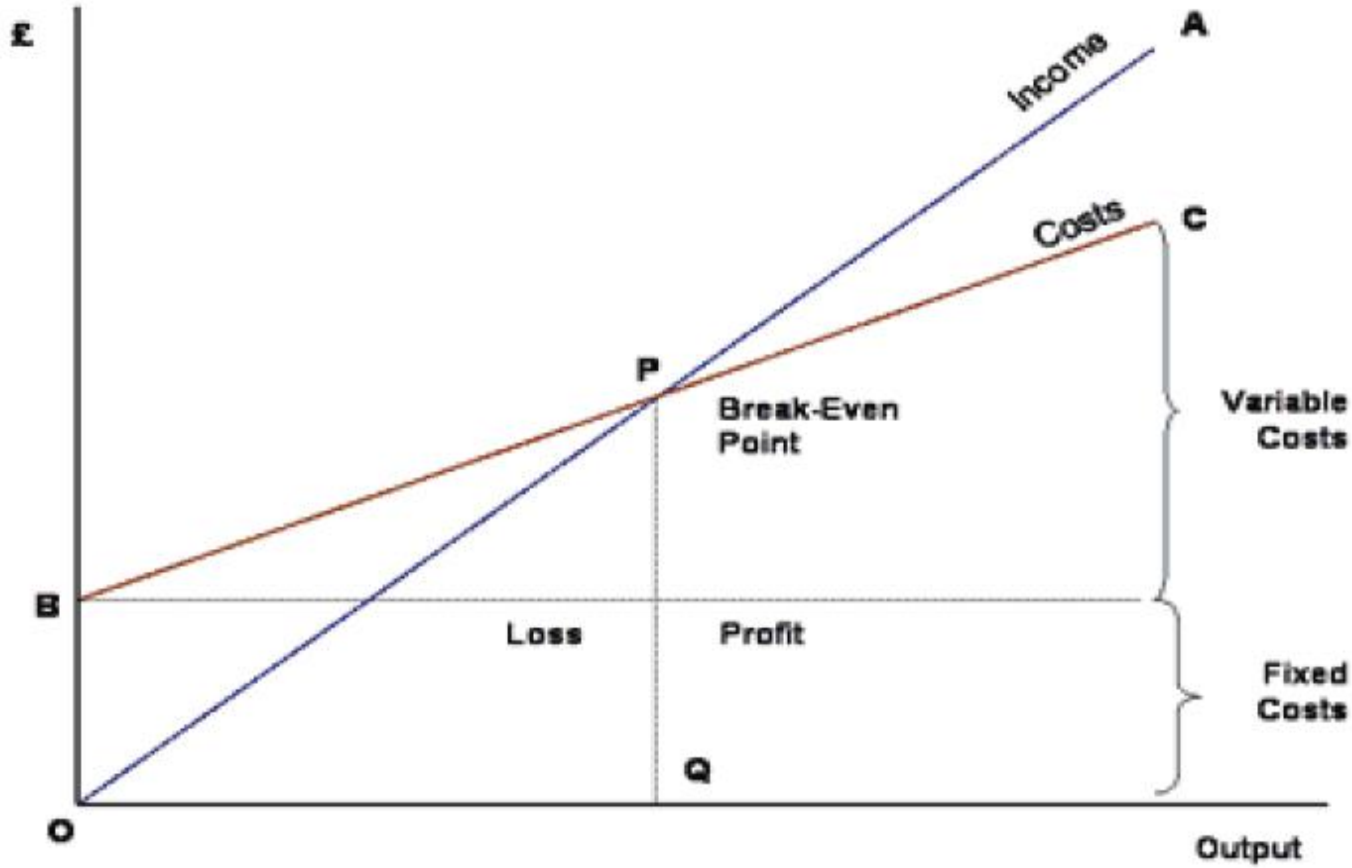
If we rearrange the formula where the breakeven is **X** then the formula look like this

$$\boxed{X = F/(P - V)}$$

Break-Event Point

This formula says that the breakeven point is where the number of sales needed to make the cost equal to the revenue.

The Break-Even Point Chart



Sample Problem # 1

Lets say you own a business selling burgers

It costs **<u>USD 7.00</u>** to make one burger That's your V or Variable cost!

You sell each burger for **<u>USD 12.50.</u>**
That's your P or price per unit!

Your cost for rent, utilities, overhead, etc... is **<u>USD 100,000</u>** per month.
That's your F or fixed cost!

Sample Problem # 1 – Solution

Given:

$V = USD\ 7, P = USD\ 12.50$

$F = USD\ 100{,}000$

- Solution:

- **$X = F / (P - V)$**
- **$X = 100{,}000 / (12.50 - 7)$ $X = 100{,}000 / (5.5)$**
- **$X = 18{,}181.818$**
- **$X \approx 18{,}182$**

To breakeven you would need to sell 18182 burgers.

Sample Problem # 2

Try out this problem for your self

- You own a lemonade stand.
- It costs you **USD 1.50** to make cup of lemonade
- You sell your lemonade for **USD 5.00**.
- It cost you **USD5,000** to rent for the space of your lemonade stand.
- How many cups of lemonade do you have to sell to breakeven?

Solve now

Sample Problem # 2 – Solution

Given:

V = USD 1.50 P = USD 5.00

F = USD 5,000

Solution:

$X = F / (P - V)$

$X = 5000 / (5 - 1.50)$

$X = 5000 / (3.5)$

$X = 1,428.57$

$X \approx 1,429$

You would need to sell 1,429 cups of lemonade to breakeven.

Limitations of BEP Analysis

- Assumes that sales prices are constant at all levels of output.
- Assumes production and sales are the same.
- Break even charts may be time consuming to prepare.
- It can only apply to a single product or single mix of products.

Conclusion

- A Breakeven Analysis is a simple tool to use to determine if you have priced your product correctly.
- A Breakeven Analysis helps you calculate how much you need to sell before you begin to make a profit. You can also see how fixed costs, price, volume, and other factors affect your net profit.